MIT Invitational - Repaired practice edition - Answer key

Division C | Remote Sensing | Repair appendix

This appendix contains authored replacements and supplied written alternatives. It does not reconstruct unpublished historical material. For the listed item IDs, use this appendix instead of the archived wording and key.

ANSWER KEY - Review only after submitting.

8. (4 points) Authored replacement for source item Question 8: Why start with a question rather than an index?

Unit: Remote sensing foundations and history.
Worked-case data: An image becomes brighter after a display stretch, but its stored reflectance values are unchanged. Has surface reflectance increased?

answer: The question determines the needed observable, wavelengths, resolution, dates, and validation; a convenient ratio may not discriminate the alternatives. (4 points).

10. (2 points) Authored replacement for source item Question 10: What does a sensor directly record in optical imaging?

Unit: Remote sensing foundations and history.
Worked-case data: An image becomes brighter after a display stretch, but its stored reflectance values are unchanged. Has surface reflectance increased?

A. Tree species with certainty

B. Human land-use intent

C. A complete climate mechanism

D. A band-dependent energy response

Correct: D

16. (2 points) Authored replacement for source item Question 16: Which sensor is active?

Unit: Sensors, radar, lidar, and illumination.
Worked-case data: A lidar pulse has round-trip time 8 microseconds. Use c = 3 x 10^8 m/s.

A. A natural-light photograph

B. Radar

C. A reflected-sunlight camera

D. A passive thermal radiometer

Correct: B